

RISKS TO DATA AND PERSONAL INFORMATION

Huge amounts of data are transmitted and stored digitally, and a lot of this data contains personal or financial information. Because of this, digital systems are targeted by criminals who try to access data so that they can use it to commit fraud or identity theft.

Threats to Data and Information

Unauthorised Access

Access to networks by users who are not permitted to access them is called **unauthorised access**.

Unauthorised users can attempt to gain access to networks directly by themselves. Alternatively, they may create software that runs thousands of times per second on devices, inputting multiple login details in order to attempt to gain access to networks with poor security.

Sometimes, devices on a network can be targeted by unauthorised users in order to be used as **botnets**. Botnets are groups of computers that have their resources used for harmful purposes, such as running and spreading malware.

Deliberate Damage by Malware

Malware (Malicious Software): Software that is created with the intention to do harm.

Malware can show messages, play sounds, delete files, or reprogram systems to perform tasks that will harm the system and the connected hardware.

Some malware, known as **ransomware**, threatens to delete a user's files or places restrictions on a user's access to software or resources until money is paid, usually to an anonymous account. These messages are usually very threatening and distressing for users. They are often written in a way that makes the user believe that they must pay quickly.

Accidental Deletion

Users can sometimes delete files or even the entire contents of a drive by mistake. This can happen if:

- They press a key on a keyboard by accident.
- They format media on the wrong storage device.
- Their device loses power unexpectedly.

Theft of Personal Data

Phishing

Phishing: The criminal activity of sending emails or having a website that is intended to trick someone into giving away personal information such as their bank account number or their computer password; this information is then used to get money or goods.

Phishing involves sending large numbers of messages that appear to be from real organisations, such as shops, banks, or charities. Phishing messages are often sent as emails. These emails ask the user to provide their information by replying to the message or following a hyperlink that opens a webpage into which the user is asked to type their personal details.

Sometimes, phishing messages are highly customised or personalised and are targeted at a smaller number of particular users. This technique is known as **spear phishing**. SMS phishing is sometimes referred to as **smishing**.

Pharming

Like phishing, **pharming** is a technique used by criminals to gain personal information and payment details from users. Criminals create fake versions of trusted websites to trick users into entering their login details, which are then used by the criminals to access users' accounts.

There are two main methods by which users are directed to a pharming site:

1. **DNS Alteration:** Internet traffic going to the real website is redirected to the fake website. Criminals do this by altering the domain name servers (DNS) to make internet traffic go to their fake site. They can also use malware to redirect web requests.
 2. **Typosquatting:** The URL of a pharming website is designed to be very similar to the URL of the real website. If a user misspells the URL when typing it into the address bar, they could go to the pharming site by mistake (e.g., visiting <http://moneybank.lk> instead of <http://moneybank.lk>).
- **Webform:** A data entry form on a web page.
 - **Internet Traffic:** Data transferred between computers connected to the internet.
 - **Domain Name Server:** A computer connected to the internet that translates domain names, such as pearson.com, into IP addresses.

Methods to Secure Data and Personal Information Online

1. Firewalls

Firewalls control the data travelling into and out of a network. They examine the network addresses and ports of the data, then compare those details to a list of rules that can be changed by network administrators. The list of rules determines what traffic should be allowed to travel into and out of the network. In this way, firewalls can prevent unauthorised access to a network and protect the network from malware.

2. Encryption

Encryption uses a key to scramble data into an unreadable form. If encrypted data is intercepted on the network, it is useless unless the interceptor has or can identify the decryption key.

3. Passwords, PINs, and Biometrics

These methods are used online to authenticate a user so that they can access an online system, such as webmail or an online bank account. When entering a password or a PIN, the characters are often **masked** (hidden) so that anyone watching the screen cannot see what is typed.

Students should ensure that their passwords follow these rules:

- More than eight characters long.
- A mix of letters, numbers, and symbols.
- A mix of uppercase and lowercase letters.
- Made up of random characters (not common words, names, or dates).
- Changed frequently.
- Something that they have not used before.

4. CAPTCHA Tests and Security Questions

- **CAPTCHA:** A computer program or system that can identify whether a user is a human or a computer. It stands for *Completely Automated Public Turing Test To Tell Computers and Humans Apart*.
- **Bot:** A computer program that can interact with systems or users.

CAPTCHA tests make sure that data is entered by a human and not by an automatic software program (a bot). They work by asking users to enter a randomly generated series of distorted

letters and numbers, select specific images (e.g., "Select all the trees"), or complete interactive tasks.

reCAPTCHA tests use extracts of text from scanned books or images. When a user solves a reCAPTCHA, their solution helps digitise books and annotate maps.

5. Anti-Malware Utilities

Anti-malware prevents malware from accessing or operating on computers. It scans computer files in **real-time** and allows users to scan specific files, folders, disks, or whole systems.

- **Anti-Virus:** Combats **viruses** (malware that uses networks to spread to connected devices via communication software like email or external devices like USBs). It constantly checks files for unique **virus definitions** (sequences of code found in viruses). If a match is found, it **quarantines** (isolates) the file in a protected area of storage so it cannot run.
- **Anti-Adware:** Detects, quarantines, and removes **adware** (software that displays unwanted advertisements).
- **Anti-Spyware:** Detects, quarantines, and removes **spyware** (software that secretly monitors and records computer data and user input, such as keyloggers that capture keystrokes).

6. Access Rights and File Permissions

- **Permissions:** Authorisation settings that provide the ability for a user or users to access files, folders, or drives.

Permissions can be set for access to files, folders, or drives, allowing users to have **read-only** access or **read-and-write** access.

7. Secure Websites (HTTPS)

- **HTTPS (Hypertext Transfer Protocol Secure):** A secure form of HTTP that authenticates payment servers and provides encryption using Secure Socket Layer (SSL) or Transport Layer Security (TLS).

HTTPS keeps communications private. Web browsers indicate a website is secure by displaying a **padlock icon** in the address bar.

8. Email Attachments and Web Links

Users must be careful when opening attachments or links. Anti-malware should be kept up to date, and students should remain alert if:

- They do not recognise the sender.
- The text is general, impersonal, or irrelevant.
- The text contains spelling or grammatical errors.
- The attached file is an executable file (such as `.exe` or `.zip` files).
- The text contains an urgent message demanding immediate action.
- The user does not recognise the URL.

9. Backup Procedures

A backup creates one or more copies of data, usually stored on an external storage device or online storage, to protect against theft, malware, flooding, fire, or power cuts.

Backup Type	Description
1. Full Backup	Creates a complete copy of all files.
2. Differential Backup	Copies all files that have changed since the last full backup .
3. Incremental Backup	Saves a copy of only the files that have changed since the last full or incremental backup .

Good backup strategies include:

- Setting up automatic backups (avoiding human error).
- Avoiding optical media (it deteriorates over time and is fragile).
- Scheduling backups for late in the evening to avoid file conflicts.
- Creating more than one copy and storing them at multiple geographical locations.
- Storing critical physical local backups inside a fireproof safe.

Online Payment Systems

1. Online Third-Party Payment Processors

Processors like PayPal or Skrill allow users to create an account to send and receive money securely using email accounts for identification.

2. Bank Cards

Bank cards allow customers to pay for items online. When making an online payment, a user must enter the:

- Card number (distinct from the bank account number)
- Expiry date

- Name on the card
- Three- or four-digit Card Security Code (CSC / CVV / CVC)

Online card systems often ask users to authenticate the transaction using secure verification steps like *Mastercard SecureCode*.

3. Contactless Cards Using NFC

Near Field Communication (NFC) transfers payment data wirelessly over short distances without requiring a PIN or user authentication. The transaction amount is strictly limited to prevent large-scale fraud if a card reader or app is used maliciously by criminals. NFC cards can be wrapped in foil to prevent weak signals from being intercepted.

Chapter Questions

- Which one of these is used to control internet traffic entering a network?**
 - A) Server
 - B) Backup
 - C) Firewall
 - D) Encryption
- Explain why CAPTCHA tests work.**
- State the purpose of pharming and phishing.**
- List three pieces of data from a bank card that a user is asked to enter when making an online payment.**
- Describe one way in which incremental backup differs from differential backup.**
- Describe how encryption secures data on a network.**
- State two methods used by an online system to authenticate a user.**
- State two methods used by criminals to get users to visit fake sites.**
- Explain one disadvantage of using online storage for backups rather than local storage.**